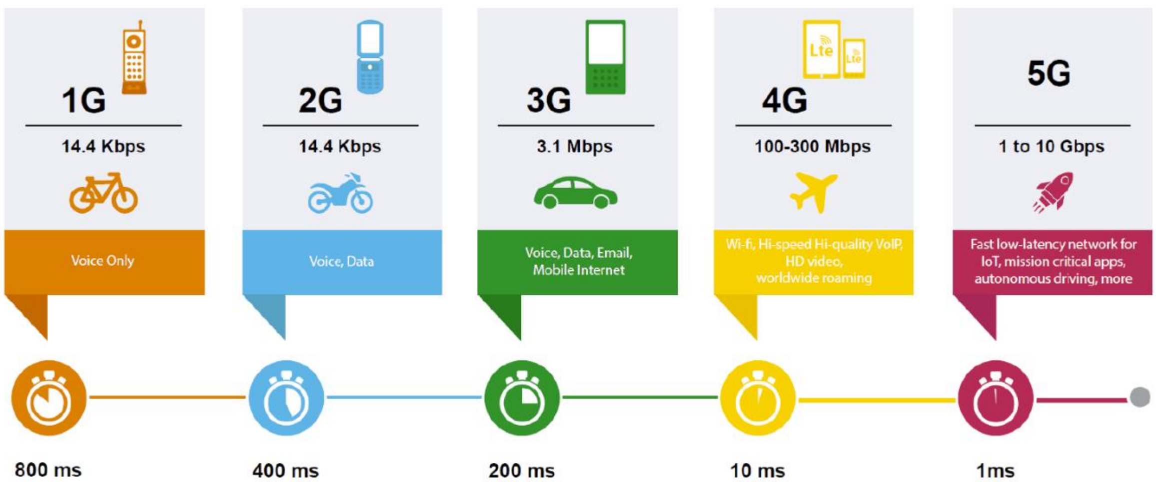
Chapter 6: Cellular Wireless Communication and Latest Trends

1. Overview of 1G, 2G, 3G, and 4G

Mobile wireless communication systems have undergone significant evolution since the launch of the first-generation mobile network in the early 1980s. Driven by the growing global demand for increased connectivity, mobile communication standards have rapidly progressed over the years to accommodate a larger number of users. This continuous advancement has led to the development of successive generations of wireless technologies for mobile communication.

Evolution of 1G to 5G



A. 1G (First Generation)

- First introduced in the 1980s and completed in the early 1990s.
- Its speed was up to 2.4 Kbps
- Voice communication only
- 1G network uses an analog signal
- Poor voice quality
- Poor battery life
- Large phone size
- No security

- Limited capacity
- A limited number of users and cell coverage
- Roaming was not possible between similar systems

B. 2G (Second Generation)

The second generation of mobile communication systems introduced a new digital technology for wireless transmission, also known as Global System for Mobile Communication (GSM).

- It was launched in Finland in the year 1991
- Based on GSM
- 2G network uses digital signals
- Its data speed was up to 64 Kbps
- It enables services such as text messages, picture messages, and MMS (multimedia message)
- It provides better quality and capacity
- These systems are unable to handle complex data such as videos
- It supports sending and receiving e-mail
- It supports web browsing
- Relatively slow data speeds for multimedia and web browsing
- Voice calls and data could not be used simultaneously on the same channel (initially).

C. 3G (Third Generation)

Third Generation (3G) wireless mobile telecommunications technology has been a key milestone in the evolution of mobile communication. It enabled significantly faster data transmission rates compared to its predecessor (2G), facilitated the rise of multimedia services, and brought global connectivity to new heights.

- Introduced in 2000s

- Data transmission speed increased from 144Kbps to 2Mbps
- Typically called smart phones and features increased its bandwidth and data transfer rates to accommodate web-based applications and audio video files
- It provides faster communication
- Send/Receive large email message
- High speed web / more security
- Video conferencing / 3D gaming
- TV streaming / Mobile TV / Phone calls
- Large capacities and broadband capabilities
- It was challenge to build the infrastructure for 3G
- High bandwidth requirement

D. 4G (Fourth Generation)

It was introduced for commercial use in Norway near the end of 2009. 4G offers today's standard services.

- Capable of providing 100Mbps to 1Gbps speed
- One of the basic term used to describe 4G is "MAGIC"
    - Mobile Multimedia
    - Anytime Anywhere
    - Global Mobility Support
    - Integrated Wireless Solution
    - Customized Personal Services
- Wireless technology that promises higher data rates
- Anytime, Anywhere as per user requirements
- More security
- High capacity
- Low cost per bit
- Battery uses is more
- Hard to implement
- Need complicated hardware

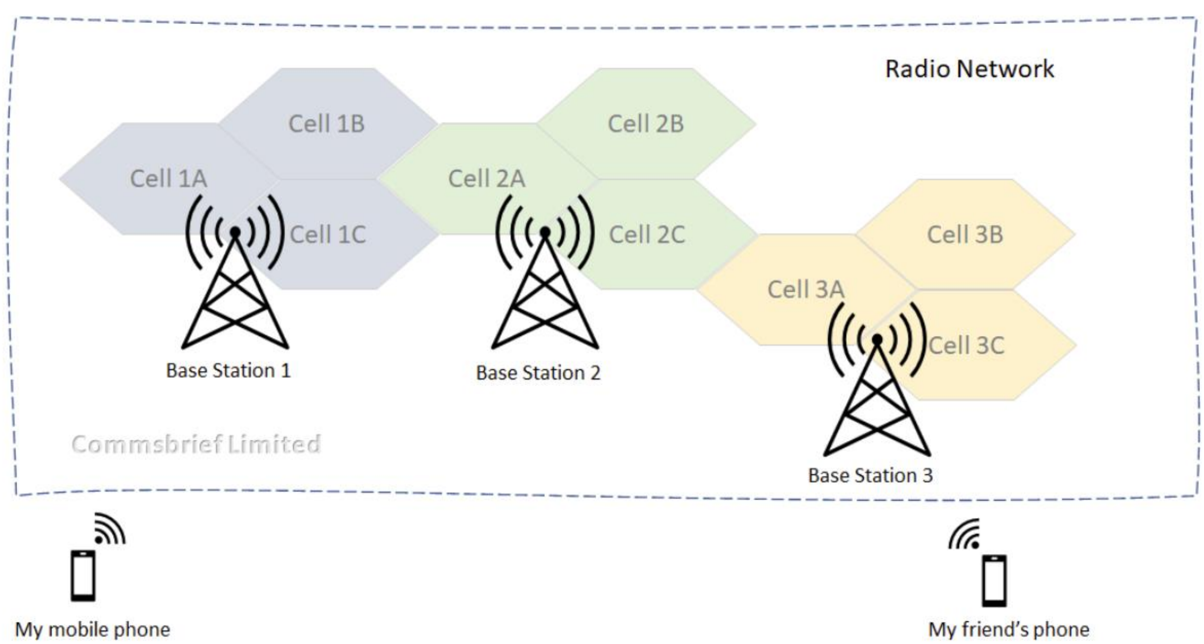
2. Cellular Technology, Fundamental Terminology: Cell, Frequency reuse, Cluster, Adjacent cell interference, Co-channel interference, Handoff strategies

Cellular Technology

Cellular technology refers to the set of wireless communication systems used to connect mobile devices and ensure data transmission through cell-structured networks. Each cell is served by a base station, which manages connectivity and enables the smooth passage of a signal from one cell to another as the user moves.

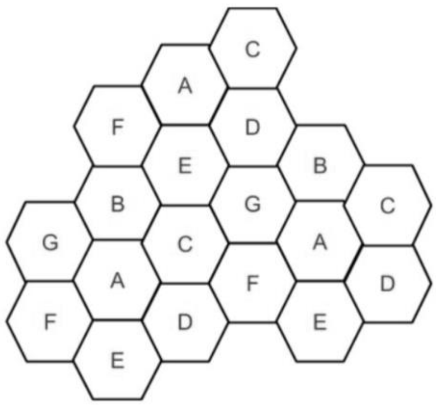
Cell

A cell is a geographical area that defines the cellular coverage zone created by the base station of a mobile network. The base station, also known as a cell tower, is equipped with transceivers that transmit and receive radio signals at licensed frequencies between the network and mobile phones. The cells in a mobile network are represented by interconnected hexagons that cover a geographical area where cellular coverage is required. In real life, the shape of a cell is determined by its range, which depends on how far the radio signals from a base station can travel before fading entirely.



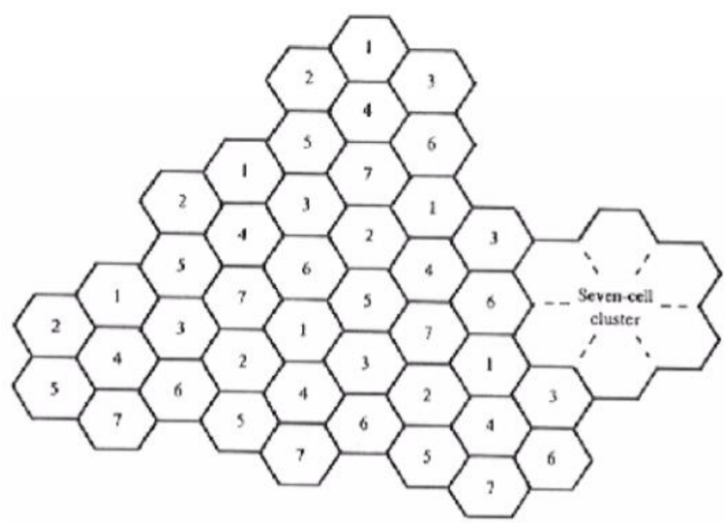
Frequency reuse

- After covering a certain distance, a radio wave gets attenuated and the signal falls below a point where it can no longer be used or cause any interference.
- A transmitter operating within a specific frequency range will have a limited coverage area.
- Beyond this coverage area, that frequency can be reused by another transmitter
- The entire network coverage area is divided into cells based on the principle of frequency reuse.
- In the figure below, cells labeled with the same letter use the same set of frequencies.



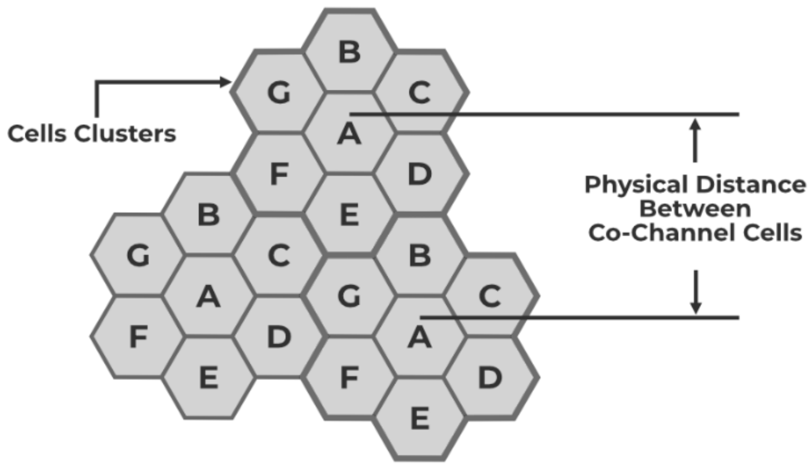
Cluster

- A set of cells can be packed in clusters such that no two similar cell are adjacent
- Possible cluster sizes are 1,3,4,7,9,12 etc.
- Frequency can only be reused outside and not within the same cluster.



Co-Channel interference

Co-channel cells are those cells that use the same frequency in a given coverage area. Interference from these cells is called co-channel interference. In co-channel interference, the cells are clustered as close together as possible to reduce the co-channel interface and provide sufficient isolation. Increasing the co-channel reuse ratio improves the transmission quality because of the smaller level of co-channel interference. An example of co-channel interference is when a radio transmitter is operating on the same frequency.

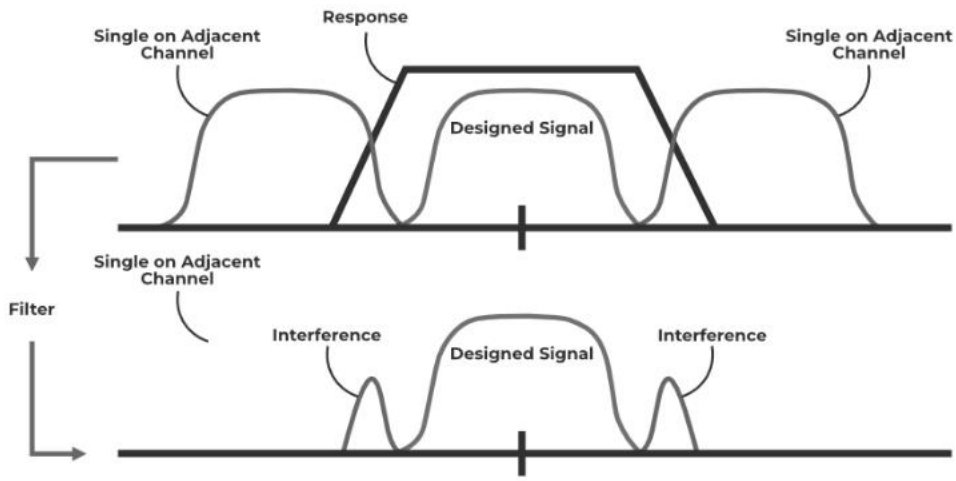


The reasons behind co-channel interference are: Bad weather conditions and poor frequency planning.

The ways we can reduce co-channel interference in cellular communication are: Proper planning & implementation, and the frequency reuse technique increases overall system capacity.

Adjacent cell interference

It is the interference caused to the signal which is adjacent in frequency to the desired signal. Imperfect receiver side filters allow the neighboring signal to mix with the actual pass band. if adjacent channel signal strength becomes strong, it will be difficult for Base Station to differentiate the actual mobile signal from the strong mobile signal.



The reasons behind adjacent channel interference are as follows:

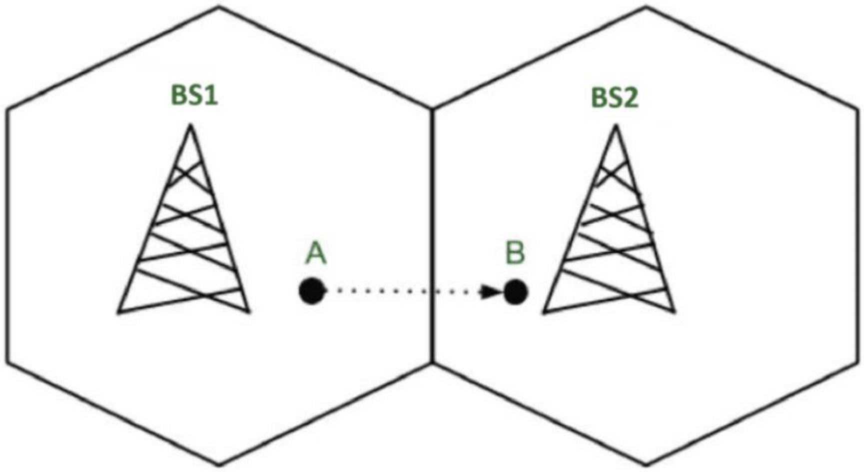
- Due to multiple channels close to each other communicating using similar frequencies.
- Irrelevant power emission from an adjacent channel.

Factors for reducing Adjacent Channel Interference are as follows:

- Proper filtering
- Careful Channel Assignments
- By managing the space between two adjacent cells, which should remain constant.

Handoff

In cellular telecommunications, the terms handover or handoff refers to the process of transferring an ongoing call or data connectivity from one Base Station to another Base Station. When a mobile moves into a different cell while the conversation is in progress then the MSC (Mobile Switching Centre) transfers the call to a new channel belonging to the new Base Station.



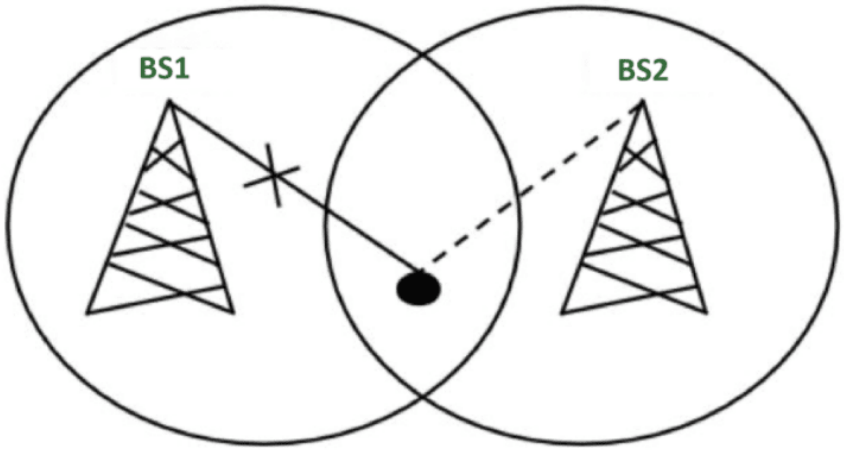
When a mobile user A moves from one cell to another cell then BSC 1 signal strength loses for the mobile User A and the signal strength of BSC 2 increases and thus ongoing calls or data connectivity for mobile users goes on without interrupting.

Types of Handoff

- **Hard handoff**
- **Soft handoff**
- **Delayed handoff**
- **Mobile-Assisted handoff**

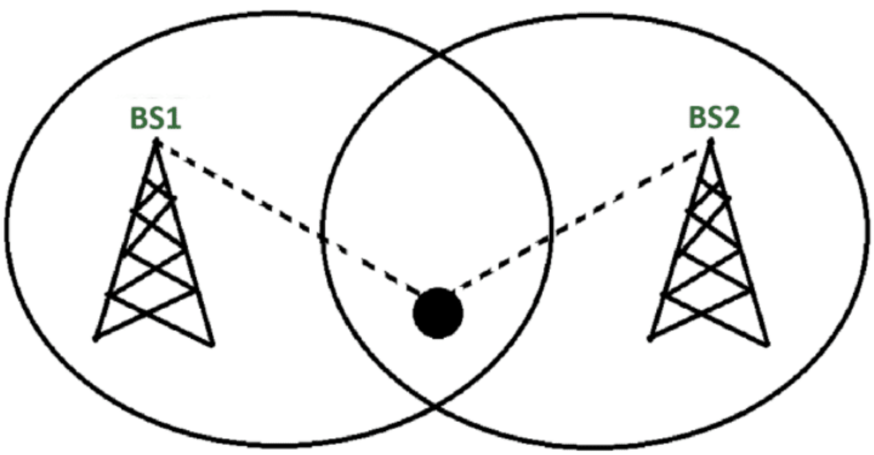
Hard handoff

When there is an actual break in the connectivity while switching from one Base Station to another Base Station. There is no burden on the Base Station and MSC because the switching takes place so quickly that it can hardly be noticed by the users. The connection quality is not that good. Hard Handoff adopted the 'break before make' policy. Hard Handoff is cheaper in cost as compared to soft handoff because only one channel needs to be active at a time. It is more efficient than soft handoff, that is why hard handoffs are widely implemented.



Soft handoff

Soft Handoff is a mechanism in which the device gets connected with two or more base stations at the same time. At least one of the links is kept when radio signals are added or removed to the Base Station. Soft Handoff adopted the 'make before break' policy.



Delayed handoff

Delayed handoff occurs when no base station is available for accepting the transfer. The call continues until the signal strength reaches a threshold, and after that, the call is dropped.

Mobile-Assisted handoff

Mobile-Assisted handoff is generally used when a mobile phone helps a base station to transfer the call to another base station with better-improvised connectivity and more signal strength. This handoff is used in TDMA technique-based GSM devices.

3. Introduction to 5G network

For more detail, please refer to the following pdf file:

5 G.pdf